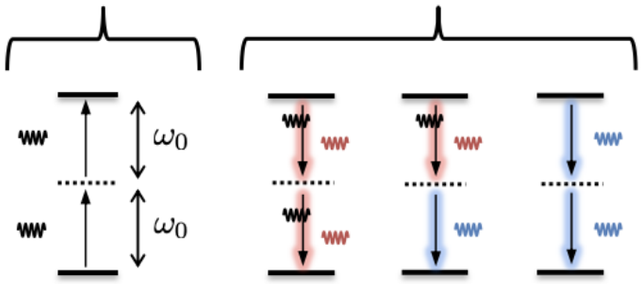
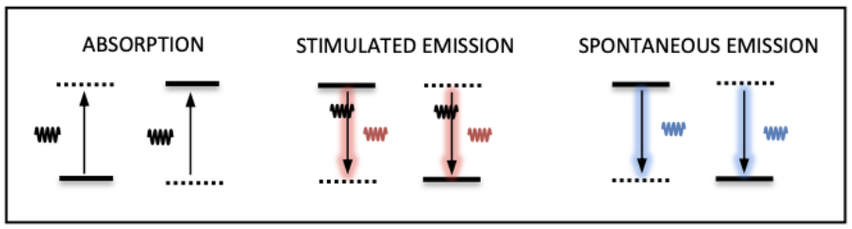
노창석 (328호)
대학원생
(334,335호)

양자 광학/정보 이론 (quantum optics/information)

양자 광학: 열린 양자 시스템

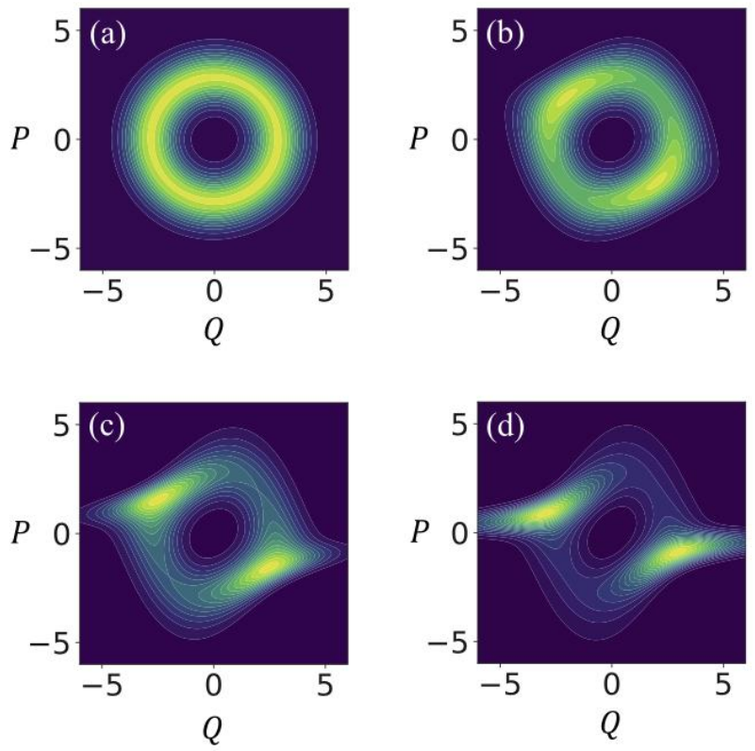
$$\frac{d}{dt}\rho(t) = \kappa_{\Downarrow}\,\mathcal{D}[\hat{a}^2]\rho(t) + \kappa_{\Uparrow}\,\mathcal{D}[\hat{a}^{\dagger 2}]\rho(t)$$





Quantum pure noise-induced transition

Quantum nonlinear oscillators



Quantum relaxation oscillations

$h(\alpha,\alpha^*) = z\,\alpha^*$

Hamiltonian
$\hat{H} = -\frac{i}{2}\,(z^*\hat{a}^2 - z\,\hat{a}^{\dagger 2})$

$h(\alpha,\alpha^*) = z_1\,\alpha^2 + z_2\,\alpha^*\alpha$

Hamiltonian
$\hat{H} = -\frac{i}{2}\,[(2\,z_1 + z_2^*)\,\hat{a} - (2z_1^* + z_2)\,\hat{a}^\dagger] - \frac{i}{2}\,(z_2^*\,\hat{a}^\dagger\hat{a}^2 - z_2\,\hat{a}^{\dagger 2}\hat{a})$

Lindblad coefficients $(N=1)$
$\eta_1 = |2z_1^* + z_2|$

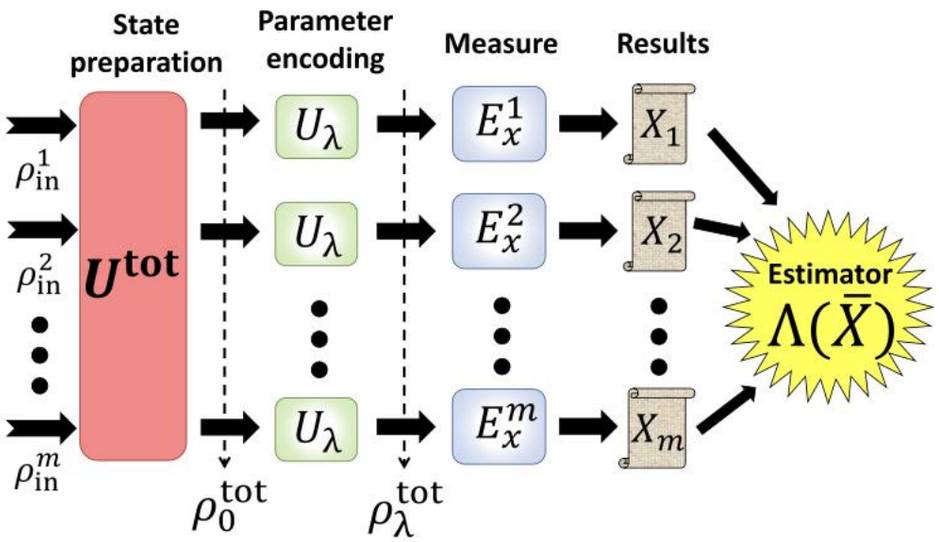
Lindblad operators
$\hat{L}_1 = \hat{a}^\dagger\hat{a} + e^{i\arg(2z_1^* + z_2)}\,\hat{a}^\dagger$

$h(\alpha,\alpha^*) = z\,\alpha^{*2}$

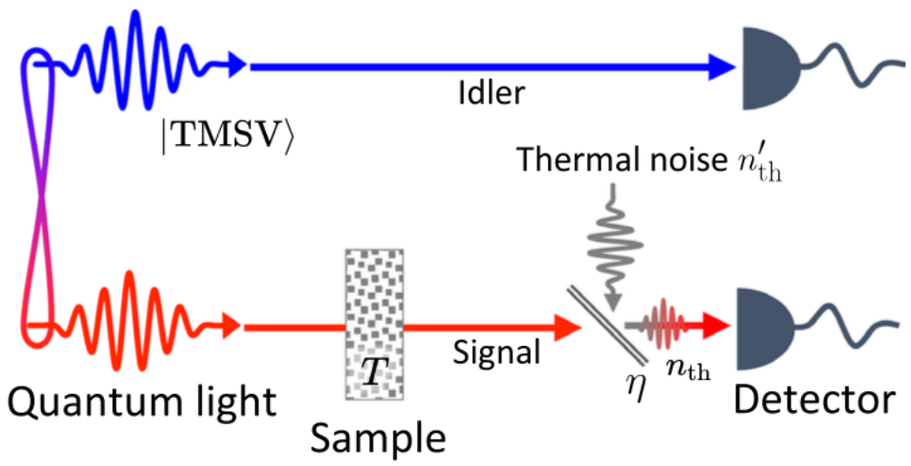
Hamiltonian
$\hat{H} = -\frac{i}{3}\,(z^*\hat{a}^3 - z\,\hat{a}^{\dagger 3})$

양자 광학/정보 이론 (quantum optics/information)

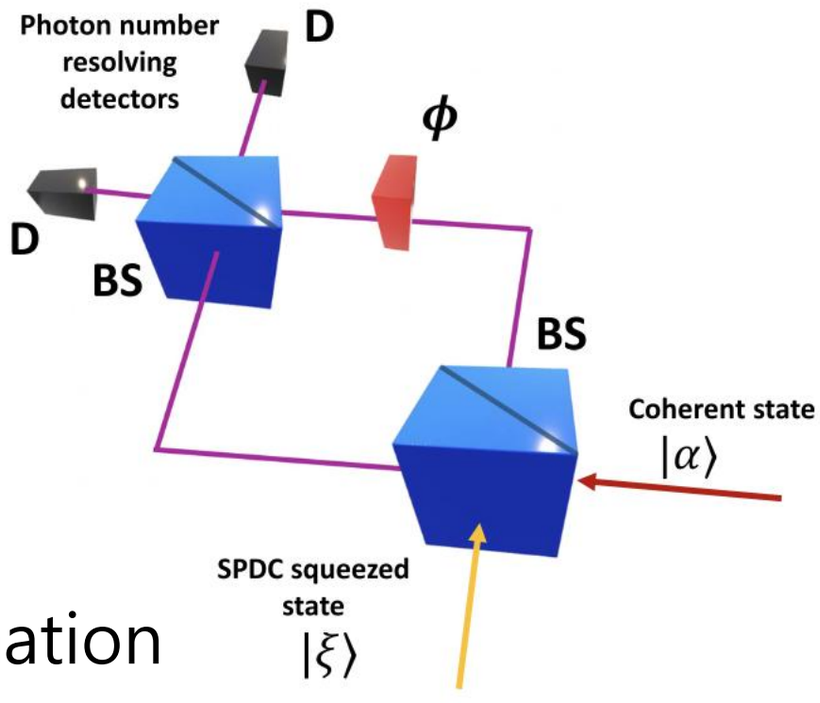
양자 정보: 양자 메트롤로지



Quantum loss sensing



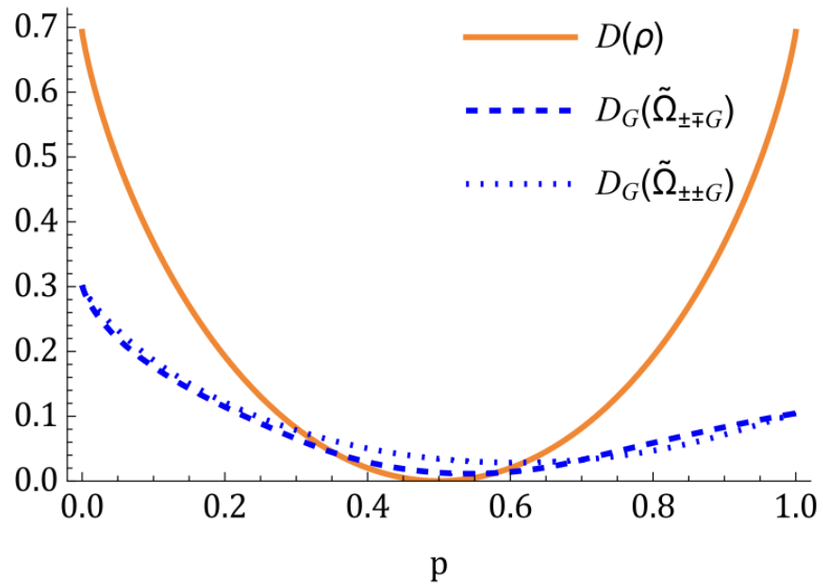
Quantum parameter estimation

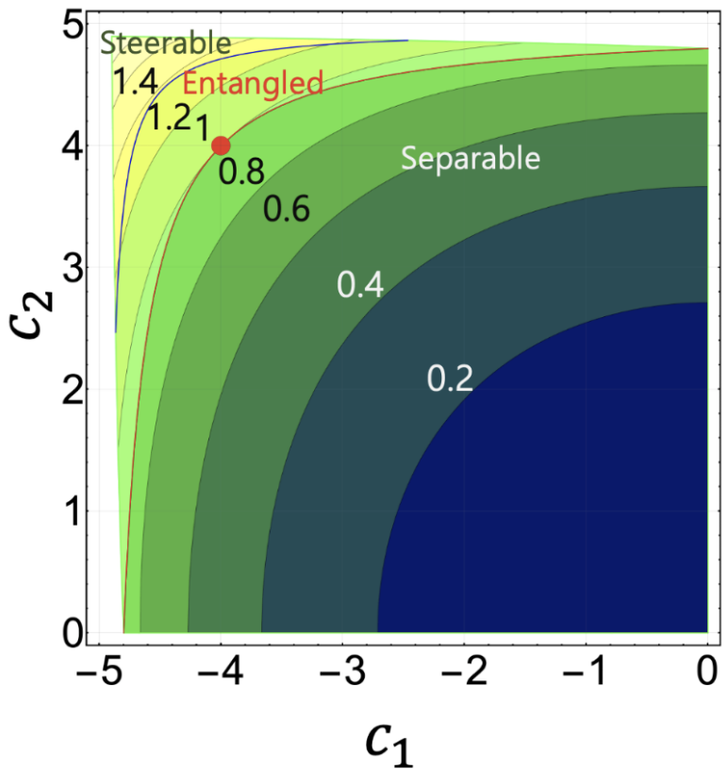


Quantum phase estimation

양자 광학/정보 이론 (quantum optics/information)

가우시안 양자 정보 이론





양자 시뮬레이션

